

Chapter 5 - A Simple Model

In this section we will start with the concept of model, and then develop the concept of statistical model. We present a simple model, the empty model, and from there begin developing the important concept of error. We also will introduce some mathematical notation for describing statistical models.

5.1 What Is a Model, and Why Would We Want One?

You may have heard people talk about statistical models. Someone may say, "I built a model based on the data," or, "Our model predicts that..." Most people who hear such statements just nod their heads, knowingly. But inside they are wondering: what does that mean?

We introduce the idea of a model by analogy. Let's say you want to estimate the area in square miles of the State of California. (Assume, for the moment, that you can't just look it up in Wikipedia!) This is not easy, because California is a large and irregularly shaped state, as pictured below.

The shape of the state California

One way to go about this would be to model the area of the state using geometric figures—specifically, rectangles and triangles. If we draw two rectangles and a triangle, and put them together in the right way, we can get something that looks somewhat like California.

The map of California next to an image in the shape of California made of geometric figures

One thing to notice about our model at the outset is that it is not very good. Most models, in fact, are pretty bad. Although the model is vaguely California-like, there are a lot of details that are not captured in the model. But, it is better than nothing! And, because we know how to calculate the area of a rectangle and the area of a triangle, if we are willing to assume the model is good enough, we can easily calculate an estimate for the area of the state.

Let's take the analogy a little further. Once we have decided to model the area of California as two rectangles and a triangle, we try to make the model fit as well as possible. To do this we adjust the size and placement of the geometric shapes. Here are a few examples:

Three maps of California overlaid with three different sizes of geometric figures

Notice that in panel c, our model seems like it will overestimate the area of California, because we've made the shapes too big. In panel b, the model looks like it will result in an underestimate. The model in panel a, however, looks pretty good. We might say that the model in panel a is the "best-fitting model" of the three.

Note that our judgment of fit comes from looking at both the area within the state that is not covered by our model (see the yellow parts sticking out in panel b), and the area not in the state that is covered by the model (see blue parts that are not overlapping with California in panel c). It is useful to think of these two areas as deviations from the model, or error.

If we write a word equation to represent this model, it might look something like this:

Area of California = Area of Geometric Figures + Other Stuff

The "other stuff" would be the deviations, both positive and negative, from the model. It is useful to think of the best-fitting model as the one that minimizes the deviations (or error), making the "other stuff" as small as possible.

Now that we have models, we can be more specific about what we mean by "other stuff." The "other stuff" becomes error from the model and we can rewrite our word equation like this:

Area of California = Area of Geometric Figures + Error

Finally, it is also important to note that our geometric model of California is a gross oversimplification of all the different attributes of California one might be interested in. Models are always like this: they oversimplify some aspects of the world, and focus only on the dimension you are most interested in.

5.2 Modeling a Distribution with a Single Number

Building on this concept of model, let's now develop what we mean by a statistical model. Whereas in the previous section we were building a model to help us estimate the area of the state of California, we now want to build a model we can use to characterize a distribution.

As you will see in subsequent chapters, statistical models are very useful. We use them to summarize distributions. We use them to make predictions about what the next observation added to a sample distribution might be. And we use them to explain variation in one variable with another. But we will start with the simplest model, which uses a single number to characterize a distribution.

At its most basic level, a statistical model can be thought of as a function that produces a predicted score for each observation in a distribution. By "function" we don't mean an R function; we mean a mathematical procedure for generating a number based on the data. The simplest models we consider generate the same predicted score for every observation in a distribution—a single number to characterize a whole distribution.

If you had to pick one number to represent an entire distribution, how would you pick it? And what would it be? Thought of in a different way: if you wanted to predict what the value of the next randomly chosen observation would be, what would be your best prediction?

Histogram of Age in MindsetMatters. The distribution ranges from about 20 to about 65, with a peak around 27 to 28 and another peak around 39 to 40.

Let's look at some distributions. What single number would you choose to represent, or model, this distribution of "Age of Housekeepers"? Why? If you wanted to predict what the next randomly chosen observation would be, what would be your prediction?

What single number would you choose to represent, or model, this distribution of "Weight of Housekeepers"? Why? If you wanted to predict what the next randomly chosen observation would be, what would be your prediction?

What single number would you choose to represent, or model, this distribution of "Predicted GPA of Students"? Why? If you wanted to predict what the next randomly chosen observation would be, what would be your prediction?

What is the best value to represent, or model, this distribution of "Race/Ethnicity of Students"? Why? If you wanted to predict what the next randomly chosen observation would be, what would be your prediction?

What is the best value to represent, or model, this distribution of "Job Status of Students"? Why? If you wanted to predict what the next randomly chosen observation would be, what would be your prediction?

For a quantitative variable whose distribution is roughly symmetrical and bell shaped, a number right in the middle might be the best-fitting model. (Remember, we aren't saying that such a simple model is a good model—just better than nothing!) If a distribution is skewed left or right, the best model might be a number toward where the middle would be if you ignored the long tail on one side or the other. For a categorical variable, the best model is generally the category that is most frequent.

Model and Error

Let's zero in on just distributions of quantitative variables. Take a look at the two distributions below for variables 1 and 2.

A faceted density histogram of the distribution of a variable named "outcome" split into two groups labeled 1 and 2. The top histogram, labeled 1, is normally distributed, centered around 100 and ranges from about 85 to 115. The bottom histogram, labeled 2, is normally distributed, centered around 100, and ranges from about 50 to 150.

What number would you choose as the best model for distribution 1 (in navy blue)? Why?

What number would you choose as the best model for distribution 2 (in sky blue)? Why?

For which distribution (1 or 2) is the model a better fit for the data? Why?

Which brings us to another important concept: once we choose a number to model a distribution (and we'll talk soon about how we do that), we can think of the variation around that number as error, just as we considered the parts of California not covered by geometric shapes as error.

If we use a single number to model the distribution of a quantitative variable, error from the model can be seen as deviations of the observed scores from that predicted number. As we just saw, a one-number model for a distribution with less spread seems to have less error, and thus a better fit, than a one-number model for a distribution with more spread. The reason for this is that the error around the model is greater for the distribution with more spread.

The idea of modeling a distribution with a single number gives us a more concrete and detailed way of thinking about our models. Whereas we thought about the California example like this:

Area Of California = Area Of Geometric Figures + Error

We can think of a statistical model like this:

DATA = MODEL + ERROR

Each data point in a distribution can be decomposed into two parts: the model (i.e., the number we are using to represent the whole distribution), and the data point's deviation from the model (the error).

5.3 The Median vs. Mean as a Model

Having developed the idea that a single number can serve as a statistical model for a distribution, we now ask: which single number should we choose? We have been talking informally about choosing a number in the middle of a symmetric, normal-shaped distribution. But now we want to get more specific.

Recall that in the previous section we defined a statistical model as a function that produces a predicted score for each observation. Armed with this definition, we can now ask: what function could we use that would generate the same predicted value for all observations in a distribution?

Median and Mean: Two Possible Functions for Generating Model Predictions

If we were trying to pick a number to model the distribution of a categorical variable, we should pick the mode; really, there isn't much choice here. If you are going to predict the value of a new

observation on a categorical variable, the prediction will have to be one of the categories and you will be wrong least often if you pick the most frequently observed category.

For a quantitative variable, statisticians typically choose one of two numbers: the median or the mean. The median is just the middle number of a distribution. Take the following distribution of five numbers:

5, 5, 5, 10, 20

The median is 5, meaning that if you sort all the numbers in order, the number in the middle is 5. You can see that the median is not affected by outliers. So, if you changed the 20 in this distribution to 20,000, the median would still be 5. (If there is an even number of values, the median is calculated as the average of the middle two values.)

To calculate the mean of this distribution, we simply add up all the numbers in the sample, and then divide by the sample size, which is 5. So, the mean of this distribution is 9. Both mean and median are indicators of where the middle of the distribution is, but they define "middle" in different ways: 5 and 9 represent very different points in this distribution.

In R, these and other statistics are very easy to find with the function `favstats()`. Create a variable called `outcome` and put in these numbers: 5, 5, 5, 10, 20. Then, run the `favstats()` function on the variable `outcome`.

If our goal is just to represent a distribution with a single number, sometimes the median is better, and sometimes the mean is better.

With the simple distribution discussed above (5, 5, 5, 10, 20), we could come up with reasons to favor both the median and the mean. In what sense might the median be a better model for this distribution? In what sense might the mean be a better model?

If you are trying to choose one number that would best predict what the next randomly sampled value might be, the median might well be better than the mean for this distribution. With only five numbers, the fact that three of them are 5 leads us to believe that the next one might be 5 as well.

On the other hand, we know nothing about the Data Generating Process (DGP) for these numbers. The fact that there are only five of them indicates that this distribution is probably not a good representation of the underlying population distribution. The population could be normal, or uniform, in which case the mean would be a better model than the median. The point is, we just don't know.

Realizing this limitation, let's look below at the distributions of several quantitative variables. For each variable, make a histogram and get the `favstats()`. Then decide which number you think would be a better model for the distribution – the median or the mean.

Note that there are two ways of asking `favstats()` or `gf_histogram()` to retrieve a variable that is inside a data frame: by using the `$` like this: `favstats(Fingers$GradePredict)` ; or by using a combination of `~` and `data =` like this: `favstats(~ GradePredict, data = Fingers)` . We prefer to use the latter version with the tilde (`~`) because it will work better with other functions we will learn about.

In general, the median may be a more meaningful summary of a distribution of data than the mean, when the distribution is skewed one way or the other. In essence, this discounts the importance of the tail of the distribution, focusing more on the part of the distribution where most people score. The mean is a good summary when the distribution is more symmetrical.

But, if our goal is to create a statistical model of the population distribution, we almost always—especially in this course—will use the mean. We shall dig in a little to see why. But first, a brief detour to see how we can add the median and mean to a histogram.

Adding Median and Mean to Histograms

You already know the R code to make a histogram. Let's add a vertical line to show where the mean is. We know from `favstats()` that the mean is 9, so we can just add a vertical line that crosses the x-axis at 9. Let's color it blue.

Try modifying this code to draw a purple line for the median of this tiny set of numbers. (The median is 5.)

You can string these commands together (using `%>%`) to put both the mean and median lines onto a histogram. (This time, we used the `mean()` and `median()` functions instead of typing in the actual numbers.)

A histogram of the distribution of outcome with a vertical line in blue showing the mean and a vertical line in purple showing the median.

Note there is a related function called `gf_hline()` that will place a horizontal line on a plot (it takes `yintercept` as an argument).

5.4 Exploring the Mean

It's pretty easy to understand how the median is the middle of a distribution, but in what sense is the mean the middle? One way to think of the mean is as the balancing point of the distribution, the point at which the things above it equal the things below it. But what does it balance? What are "the things" that are equal on both sides of the mean?

You might think that the values below the mean balance with the values above the mean. Let's try that. Does $5+5+5 = 10+20$? No, 15 does not equal 30. A bunch of smaller values, what we find below the mean, is not going to balance a bunch of larger values (the ones above the mean). So what does the mean balance?

Here it helps to think about each score's deviation from the mean, which would be negative for scores below the mean and positive for scores above. In our example, each of the 5s are 4 units below the mean of 9, which is a deviation of -4. If you think of it this way, the sum of deviations below the mean (-12) balances out the sum of deviations above the mean (+1 and +11, or +12).

We will also call these differences residuals. The word deviation is specific to differences above and below the mean but residual more generally means differences above and below any model of the distribution, which could be mean, median, mode, etc.

A dotplot with the mean represented as a vertical line. Each dot is connected to the mean line by a horizontal line to represent the distance of each value from the mean. It shows that the sum of those distances is zero.

It turns out that no number other than the mean (not 8, not 8.5, not 9.1!) will perfectly balance the residuals above the mean with those below the mean. Whereas the magnitude of a score—especially an outlier—won't necessarily affect the median, it will affect the mean because the large residual from an outlier has to be balanced with the residuals from the other data points. Every value gets taken into account when calculating the mean.

Remember we talked about finding some simple shapes that "fit" the more detailed shape of California the best? We wanted to find shapes that were not too big and not too small, shapes that would minimize the error around the model, defined as the parts of California that were not covered by the model, and the parts of the model that covered stuff not in California.

The mean is a model that is not too big and not too small. The mean is pulled in both directions (larger and smaller) at once and settles right in the middle. The mean is the number that balances the residuals above and below it, yielding the same amount of error above it as below it. It's kind of amazing that this procedure of adding up all the numbers and dividing by the number of numbers results in this balancing point.

Thinking about the mean in this way also helps us think about $\text{DATA} = \text{MODEL} + \text{ERROR}$ in a more specific way. If the mean is the model, each data point can now be thought of as the sum of the model (9 in our outcome variable) plus its residual from the model. So 20 can be decomposed into the model part (9) and the error from the model (+11). And 5 can be decomposed into 9 (model) and -4 (error).

5.5 Fitting the Empty Model

This section might seem overly complicated at first, but it will provide the foundation for understanding subsequent concepts. The simple model we have started with—using the mean to model the distribution of a quantitative variable—is sometimes called the empty model or null model. In the context of students' thumb lengths, we could write the empty model with a word equation like this.

Thumb = Mean + Error

Note that the model is "empty" because it doesn't have any explanatory variables in it yet. The empty model does not explain any variation; it merely reveals the variation in the outcome variable (the Error) that could be explained. This empty model will serve as a sort of baseline model to which we can compare more complex models in the future.

If the mean is our model, then fitting the model to data simply means calculating the mean of the distribution, which we could do using `favstats()` .

It balances the error on each side so it has equal chance of being too low as of being too high. The mean (60.1) is a number that is neither too big nor too small of a guess. The mean perfectly balances the residuals, a way of measuring error, above and below the mean. Note in this distribution, which happens to be fairly symmetrical and normal, the mean and median make similar predictions (around 60).

When we fit a model we are finding the particular number that minimizes error the most; this is what we mean by the "best-fitting" model. The mean of a distribution is one such number because it balances the residuals. In later chapters, we will expand more on why this value is the best-fitting one.

It's easy to fit the empty model—it's just the mean (60.1 in this case). But later you will learn to fit more complex models to your data. We are going to teach you a way to fit models in R that you can use now for fitting the empty model, but that will also work later for fitting more complex models.

The R function we are going to use is `lm()` , which stands for "linear model." (We'll say more about why it's called that in a later chapter.) Here's the code we use to fit the empty model, followed by the output.

```
lm(Thumb ~ NULL, data = Fingers)
```

Call:

```
lm(formula = Thumb ~ NULL, data = Fingers)
```

Coefficients:

(Intercept)

60.1

Although the output seems a little strange, with words like "Coefficients" and "Intercept," the `lm()` function does give you back the mean of the distribution (60.1), as expected. `lm()` thus "fits" the empty model in that it finds the best-fitting number for our model. The word "NULL" is another word for "empty" (as in "empty model.")

It will be helpful to save the results of this model fit in an R object. Here's code that uses `lm()` to fit the empty model, then saves the results in an R object called `empty_model` :

```
empty_model <- lm(Thumb ~ NULL, data = Fingers)
```

If you want to see the contents of the model, you can just type the name of the R object where you saved it (i.e., just type `empty_model`). Try it in the code block below.

When we save the result of `lm()` as `empty_model` we are saving a new type of R object called a model, which is different from data frames or vectors. When R saves a model it saves all sorts of information inside the object, which we won't get into here. But some functions we will learn about specifically take models as their input.

One such function is `gf_model()` , which lets us overlay a model (e.g., `empty_model`) onto various types of plots, including histograms, box plots, jitter plots, and scatter plots.

For example, here is how to chain the `empty_model` predictions (using the pipe operator, `%>%`) onto a histogram of thumb lengths.

```
gf_histogram(~ Thumb, data = Fingers) %>%  
gf_model(empty_model)
```

A histogram of the distribution of thumb with a vertical line in blue showing the mean.

As we now know, the empty model predicts the mean thumb length for everyone. This model prediction is represented by the single blue vertical line right at the average thumb length of 60.1 mm. Let's try adding `gf_model(empty_model)` to some other plots below.

Try running the code block below and take a look at the resulting graphs. Then use the pipe operator (`%>%`) to add the empty model prediction to both the scatter plot of thumb lengths by height and the jitter plot of thumb lengths by gender. (The empty model has already been fit and saved as `empty_model` .)

Scatter plot of Thumb ~ Height Jitter plot of Thumb ~ Gender

On the left, a scatter plot of Thumb by Height in Fingers with the empty model overlaid as a horizontal line.

On the right, a jitter plot of Thumb by Gender in Fingers with the empty model overlaid as a horizontal line.

When we overlaid the empty model prediction on the histogram of Thumb , it was represented as a vertical line because Thumb was on the x-axis. With the scatter plot and jitter plot above, Thumb is on the y-axis. For this reason, the empty model prediction (i.e., the mean of Thumb) is represented as a horizontal line, right at the mean of Thumb .

The empty model prediction on the scatter plot (on the left, above), as represented by the horizontal line, indicates that regardless of how tall a student is, the empty model will assign them a predicted thumb length of 60.1 mm.

What does the horizontal line on the jitter plot (on the right in the figure above) mean?

No matter what gender you are, the model predicts that you have a 60.1mm length thumb
No matter what the sex of a student is (female or male), the empty model prediction of their thumb length is 60.1 mm. Note: this is not to say that females and males don't differ in their thumb lengths, but only that the empty model prediction does not take gender into account.

The Mean as an Estimate

If it seems like we're making a big deal about having calculated the mean, you're right! The mean is a concept that will come up throughout the course and your understanding of it and its uses will deepen over time. One point worth making now, however, is that the goal of statistics is to understand the Data Generating Process (DGP). The mean of the data distribution is an estimate of the mean of the population that results from the DGP, which is why the numbers generated by `lm()` are called “estimates” (or sometimes, “coefficients”).

The mean may not be a very good estimate—after all, it is only based on a small amount of data—but it's the best one we can come up with based on the available data. The mean also is considered an unbiased estimate, meaning that it is just as likely to be too high as it is too low.

5.6 Generating Predictions From the Empty Model

The mean of the sample distribution is our best, unbiased estimate of the mean of the population. For this reason, we use the mean as our model for the population. And, if we want to predict what the next randomly sampled observation might be, without any other information, we would use the mean.

In data analysis, we also use the word “predict” in another sense and for another purpose. Using our `Fingers` data, we found the mean thumb length to be 60.1 mm. So, if we were predicting a new student's thumb length, we'd go with 60.1 mm. But if we take the mean and look backwards at the data we have already collected, we could also generate a “predicted” thumb length for each of the data points we already have. This prediction answers the question: what would our model have predicted this thumb length to be if we hadn't collected the data?

There's an R function called `predict()` that will actually do this for you. Here's how we can use it to generate the predicted thumb lengths for each of the 157 students in the `Fingers` data frame.

You may be wondering: why would we want to create predicted thumb lengths for these 157 students when we already know their actual thumb lengths? We will go into this a lot more in the next chapter, but briefly, the reason is so we can get a sense of how far off the model

predictions are from the actual data. In other words, it gives us a rough idea of how much error there is around the model predictions, i.e., how well our model fits our current data.

In order to use these predicted scores as a way of seeing how much error there is, we first need to save the prediction for each student in the dataset. When there is only one prediction for everyone, as with the empty model, it seems like overkill to save the predictions.

But as we go, we'll start to appreciate how useful it is to save the individual predicted scores. For example, if we save the predicted score for each student in a new variable called `Predict`, we can then subtract each student's actual thumb length from their predicted thumb length to see how far off the prediction is for each student.

In the code window below, use the `predict()` function to save the predicted thumb lengths for each of the 157 students as a new variable in the `Fingers` dataset. We've added some code to overlay these predictions onto a scatter plot we've looked at before (`Thumb` by `Height`).

A snippet of the `Fingers` data frame A scatter plot of `Thumb` by `Height`

```
Thumb Predict
1 66.00 60.10366
2 64.00 60.10366
3 56.00 60.10366
4 58.42 60.10366
5 74.00 60.10366
6 60.00 60.10366
```

Scatter plot of `Thumb` predicted by `Height` in `Fingers` with the empty model overlaid as point predictions for each data point. The predictions are aligned in a horizontal line.

As we can see from the sample rows from the `Fingers` data frame (on the left), every student, regardless of their thumb length, is given the same model prediction (60.1). This is because in the empty model, there is only one prediction, and that is the mean.

In the scatter plot, we might see a relationship between `Thumb` and `Height` in the data. But the empty model ignores all that and just predicts the same thumb length for every student, which is why the predictions (`Fingers$Predict`) form a straight horizontal line. (Later we will learn how to adjust predictions based on explanatory variables such as gender or height).

5.7 Thinking About Error

We have developed the idea of the mean being the simplest (or empty) model of the distribution of a quantitative variable, represented in this word equation:

$$\text{DATA} = \text{MEAN} + \text{ERROR}$$

If this is true, then we can calculate error in our dataset by just moving components of this equation around to get the formula:

$$\text{ERROR} = \text{DATA} - \text{MEAN}$$

Using this formula, if someone has a thumb length larger than the mean (e.g., 62 versus a mean of 60.1), then their error is a positive number (in this case, nearly +2). If they have a thumb length lower than the mean (e.g., 58) then we can calculate their error as a negative number (e.g. about -2).

We generally call the error calculated this way as the residual. Now that you know how to generate predictions, we'll refine our definition of the residual to be the difference between our model's prediction and an actual observed score. The word residual should evoke the stuff that remains because the residual is the leftover variation from our data once we take out the model.

To find these errors (or residuals) you can just subtract the mean from each data point. In R we could just run this code to get the residuals:

```
Fingers$Thumb - Fingers$Predict
```

If we run the code, R will calculate the 157 residuals, but it won't save them unless we tell it to do so. Modify the code in the window below to save the residuals in a new variable in `Fingers` called `Resid`. (Note that the variable `Predict` already exists in the `Fingers` data frame).

```
Thumb Predict  Resid
1  52 60.10366 -8.103662
2  56 60.10366 -4.103662
3  64 60.10366  3.896338
4  70 60.10366  9.896338
5  66 60.10366  5.896338
6  62 60.10366  1.896338
```

These residuals (or "leftovers") are so important in modeling that there is an even easier way to get them in R. The function `resid()`, when given a model (e.g., `empty_model`) will return all the residuals from the predictions of the model.

Notice that the values for `Resid` and `EasyResid` are the same for each row in the dataset. We will generally use the `resid()` function from now on, just because it's easier, but we want you to know what the `resid()` function is doing behind the scenes.

Below we have plotted a few of the residuals from the `Fingers` dataset on the `Thumb` by `Height` scatter plot. Visually, the residuals can be thought of as the vertical distance between the data (the students' actual thumb lengths) and the model's predicted thumb length (60.1).

Note that sometimes the residuals are negative (extending below the empty model) and sometimes positive (above the empty model). Because the empty model is the mean, we know that these residuals are perfectly balanced across the full dataset of 157 students.

Scatter plot of Thumb predicted by Height from the Fingers data frame, with the empty model overlaid as a horizontal line. A few data points above and below the model are highlighted and connected to the model by vertical lines.

Distribution of Residuals

Below we've plotted histograms of the three variables: Thumb, Predict, and Resid .

A histogram of the distribution of Thumb on the left. A histogram of the distribution of Predict in the middle. There's no variation in this distribution. A histogram of the distribution of Resid on the right. The distribution of Thumb and the distribution of Resid have the exact same shape, but different means.

Examine the distributions above.

What does the distribution of Thumb (data) look like?

What about the distribution of Predict (model)?

Finally, what about the distribution of Resid (error)?

What is different about the distributions of data and error? What is similar?

The distributions of the data and the residuals have the same shape. But the numbers on the x-axis differ across the two distributions. The distribution of Thumb is centered at the mean (60.1), whereas the distribution of Resid is centered at 0. Data that are smaller than the mean (such as a thumb length of 50) have negative residuals (-10) but data that are larger than the mean (such as 70) have positive residuals (10).

R will sometimes give you outputs in scientific notation. The $-1.47\text{e-}14$ is equivalent to which indicates that this is a number very close to zero (the -14 meaning that the decimal point is shifted to the left 14 places)! Whenever you see this scientific notation with a large negative exponent after the "e", you can just read it as "zero," or pretty close to zero.

The residuals (or error) around the mean always sum to 0. The mean of the errors will also always be 0, because 0 divided by n equals 0. (R will not always report the sum as exactly 0 because of computer hardware limitations but it will be close enough to 0.)

5.8 Venturing Into the World of Mathematical Notation

Up to now we have been representing our models with word equations: Thumb = Gender + Error; and so on. All of these follow the form: DATA = MODEL + ERROR. But writing out all these words will get tiresome as we write more models, which is why statisticians have developed mathematical notation to represent these relationships. We aren't going to teach lots

of notation in this course, but we are going to teach some. It's a great way to impress your friends and employers, and also it's a useful way to help you think about statistics.

Representing the Mean

You probably learned how to calculate a mean (or average) in elementary school: take a group of numbers, add them together, and then divide by the number of numbers in your group. We could represent this calculation like this:

$$\text{Mean} = \frac{\text{Sum of all the numbers}}{\text{Number of numbers}}$$

As you can see, in the expression above we used informal notation, similar to the word equations we have been using up to this point. Assuming you know that in a fraction the fraction bar can be read as "divided by" (such that a/b means "a divided by b") then you can see clearly what the expression above represents: the sum of a group of numbers divided by the number of numbers. Simple.

If we rewrite the informal expression in mathematical notation we might write it like this—which we admit—looks a lot more complicated:

$$\bar{Y} = \frac{\sum_{i=1}^n Y_i}{n}$$

Take a deep breath, no need to panic. It's really pretty simple. Let's unpack this equation piece by piece:

We use the letter $\bar{Y}$ to represent the outcome variable, and we will do this throughout the course.

Putting the bar over ($\bar{Y}$) simply means the "mean of Y." This is pronounced "Y bar."

The Greek capital letter Σ (pronounced "sigma") means sum, and should be read as "the sum of..."

n is the letter used to represent the size of a sample.

And finally, we use the subscript i to represent each individual observation in our sample, starting from observation and counting up to observation.

So, in the equation above, the expression to the right of the equal sign could be read as: the sum of all individual Ys from 1 to n, divided by n (the number of observations).

This seems like a lot of notational baggage just to represent the mean of a sample. But, it has components that will become more and more useful as you work through the course. And, it resolves ambiguities that could come back to haunt us later if we just use informal word equations. For example, if we say "sum of all the numbers," which numbers, exactly, are we referring to? The outcome variable? The explanatory variable? Using the notation Y makes this clear.

As a minor detour, let's go back to one of the properties of the mean that we previously discovered using R: if you add up the deviations of each score in a distribution from the mean of the distribution, you will get 0. We could use notation to represent that idea like this:

$$\sum_{i=1}^n (Y_i - \bar{Y}) = 0$$

Two Important Points About Notation

There are two things you need to understand about mathematical notation—things that most students never fully appreciate but mathematicians know well. First, there is no single correct way to use mathematical notation. So when you pick up multiple textbooks in statistics, they often will have different formulas for things like the mean and the standard deviation, as well as for everything else. You need to be flexible and not get flustered if you see different expressions that basically mean the same thing.

At some point, people get tired of writing the whole "i = 1 to n" part, and just agree to drop it because we all know what this means without it: sum up all the individual Ys and divide by n. This is all part of the living world of mathematical notation.

Because teachers want you to use notation consistently, they often will tell students to use the exact notation used in the particular textbook being used. But don't get fooled by this: other books will write things differently. You need to be able to look at different versions of an expression or formula and see that they are really the same thing. (But yes, you also probably need to remember the one version your teacher expects you to use.)

The other thing about mathematical notation is that there are two different ways of interpreting it. Most students look at notation (e.g., the equation for the mean, above) as a step-by-step recipe for how to produce an answer. So, to find the mean, take all the numbers, add them up, and then divide by n. But there's a far more powerful way to look at mathematical notation: it is a representation of a quantity or relationship.

To take a simple example, the notation 2+3 could be seen as a recipe: take the number 2, add 3 more, and you'll get the answer. But it also could be seen as a quantity in and of itself, the quantity (2+3).

You may not think it makes much difference, but what if we give you instead the expression $(x+3)$. If you see this as a recipe, and if you don't know what is, you are stuck. But if you look at it as representing the quantity $(x+3)$ then you don't need to know the value of x . You can still think of it as a quantity that is 3 greater than x , regardless of whether you can compute it.

In statistics, if you think of notation in this second way, as a representation of quantities and relationships, you will end up learning and understanding more. So, when you see an expression that represents the mean, don't think of it as a handy set of instructions for how to calculate the mean. And don't start calculating the mean. You really don't need to do that anyway; your computer will calculate the mean for you (think `favstats()`), and you don't need to tell it how!

Instead, think of it as a representation of an important relationship that defines what the mean is. When you see an equation, think about what it represents, look for patterns, and so on.

5.9 DATA = MODEL + ERROR: Notation

Now let's see how mathematical notation is used to represent the simple (empty) model we introduced before.

Thumb = Mean + Error

There are some real advantages to rewriting this statement in mathematical notation. Here's one form this notation might take:

$$Y_i = \bar{Y} + e_i$$

This equation literally represents what we showed above in our word equation. It tells us that each value of Thumb in our dataset (Y_i) can be seen as the sum of two parts: the mean of all values of Y ($\bar{Y}$, which is the empty model prediction), and its residual from that mean (e_i , or error). If we add these two numbers together (Mean + Error) for a specific score, we will get the original score. Very simple, very concrete.

Notation for the General Linear Model

This notation for the mean ($\bar{Y}$) works well enough for the empty model. But it's not going to help us build more complex models later. To prepare for that eventuality, we will introduce a more general notation, referred to as the General Linear Model (GLM). In GLM notation, the empty model is represented like this:

$$Y_i = b_0 + e_i$$

Based on what you have learned about notation so far, what do you think the b_0 might represent in the equation above?

This is a more general version of the equation above, in which we have substituted b_0 (we read this as "b sub o") for the mean, $\bar{Y}$. This won't make much sense right now, but later it will help us add complexity to our model (with b_1 , b_2 , and so forth). The main thing to know for now is that b_0 can represent the mean, as it does in the empty model, but it won't always represent the mean.

$$\underbrace{Y_i}_{\text{Thumb}} = \underbrace{b_0}_{\text{Predict}} + \underbrace{e_i}_{\text{Resid}}$$

Indeed, this flexibility is what makes the General Linear Model general. Whenever you see a GLM model statement, you should think carefully about what, in the particular situation, each symbol represents.

Statistics and Parameters

Now is a good time to remember that our goal in exploring distributions of data is to find out about the DGP. Our goal in constructing statistical models is the same: we estimate models based on data in order to make inferences about the population and the DGP.

With our data, we can calculate the exact mean of the distribution, and the exact size of the errors. When we do this, we are calculating a statistic. A statistic is anything you can compute to summarize something about your data; the mean is our first example of a statistic.

But we can't calculate the mean of the population; the population distribution is unknown. Instead we use the mean we calculate from our data as an estimate of the mean of the population—the distribution from which our data were sampled.

The mean of the population is an example of a parameter. A parameter is a number that summarizes something about a population. Whereas statistics are computed, parameters are estimated. We use statistics as estimates because we don't generally know what the true parameter is.

When we use the mean as a model, why do we call it a "parameter estimate"?

- A. Because we can't calculate the mean of a sample, we must estimate it.
- B. Because we can't calculate the mean of the DGP, we must estimate it.
- C. Because whenever we use R to calculate something, we call it an estimate.
- D. Because the formula for the mean is very complicated, we call it an estimate.

What is the difference between a model of data and a model of the DGP?

- A. What we want is a good model of the DGP. Because we can't develop one directly, we must instead develop a model based on sample data.
- B. Our certainty about each model's accuracy is different. We can be certain about our sample model statistics, but never truly know the parameters in the model of the DGP.
- C. Both of the above are differences.

D. Neither of the above are differences.

Sometimes students think that the main goal of statistics is to calculate a correct answer. But statistics isn't mostly about calculation. It is mostly a way of thinking about how to interpret those calculations. In statistics, understanding what you are trying to calculate can be just as important as the calculations themselves.

Notation is one way we keep our thinking straight about what we are trying to calculate, and what the results of our calculations mean. Because the distinction between statistics (or estimators) and parameters is so critical, we use different notation to distinguish them.

If we want to represent the mean calculated from data, we typically use the notation $\bar{Y}$ (or, sometimes, $\bar{X}$). To represent the mean of the population, we typically use the Greek letter μ (pronounced "mew").

The same distinction shows up in the notation of the General Linear Model. The empty model we have discussed so far, which is calculated from data, is written like this (as you know):

$$Y_i = b_0 + e_i$$

The model of the DGP that we are trying to estimate when we fit the empty model is represented like this:

$$Y_i = \beta_0 + \epsilon_i$$

Note that in this model of the population we have replaced the estimators b_0 and e_i with the Greek letters β_0 (pronounced "beta sub o") and ϵ_i (pronounced "epsilon sub i"). b_0 is the estimator for β_0 , which is used to represent the mean of the population; and e_i is the estimator for ϵ_i .

Whenever you see Greek letters you can be pretty sure we are talking about parameters of the population. Roman letters are generally used to represent estimators calculated from data. Models are often referred to by the number of parameters that are estimated. The empty model can be referred to as a "one-parameter model" because only one parameter (β_0) is estimated.

```
empty_model <- lm(Thumb ~ NULL, data = Fingers)
empty_model
```

Call:

```
lm(formula = Thumb ~ NULL, data = Fingers)
```

Coefficients:

(Intercept)

60.1

When we ran this R code to fit the empty model to our data for the thumb lengths from the full Fingers data set, what was the number 60.1? (Check all that apply.)

- A. A statistic
- B. A parameter
- C. A parameter estimate

As it turns out, in the absence of other information about the objects being studied, the mean of our sample is the best estimate we have of the actual mean of the population. It is equally likely to be too high as it is too low, making it an unbiased estimator of the parameter.

Because it is our best guess of what the population parameter is, it is the best predictor we have of the value of a subsequent observation. While it will certainly be wrong, the mean will do a better job than any other number.

Let's say we run a study and calculate a parameter estimate, $b_0 = 25$. We decide to run the study again, and this time get the same parameter estimate of $b_0 = 25$. Based on these two studies, what do we know about the population parameter β_0 ?

- A. We know the true population parameter $\beta_0 = 25$, because both samples confirmed this.
- B. We know the true population parameter is probably 25, but we'd have to do the study more than twice to be sure.
- C. We know that 25 is our best estimate of β_0 , but we can never be sure of the true population parameter.